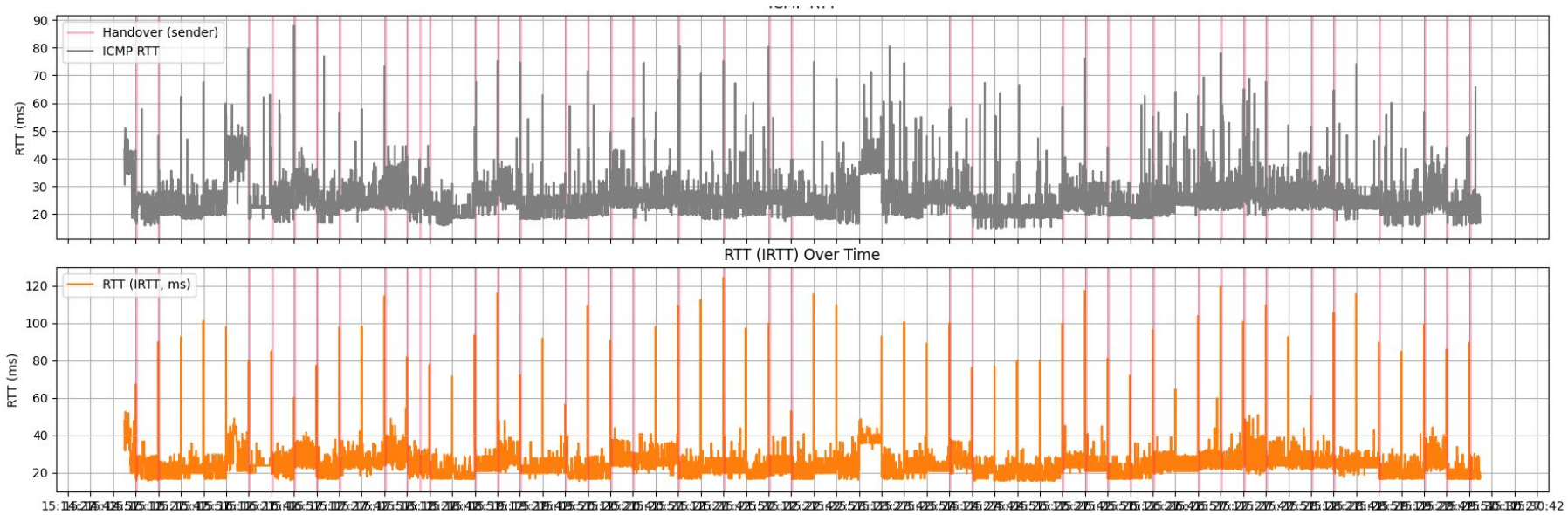
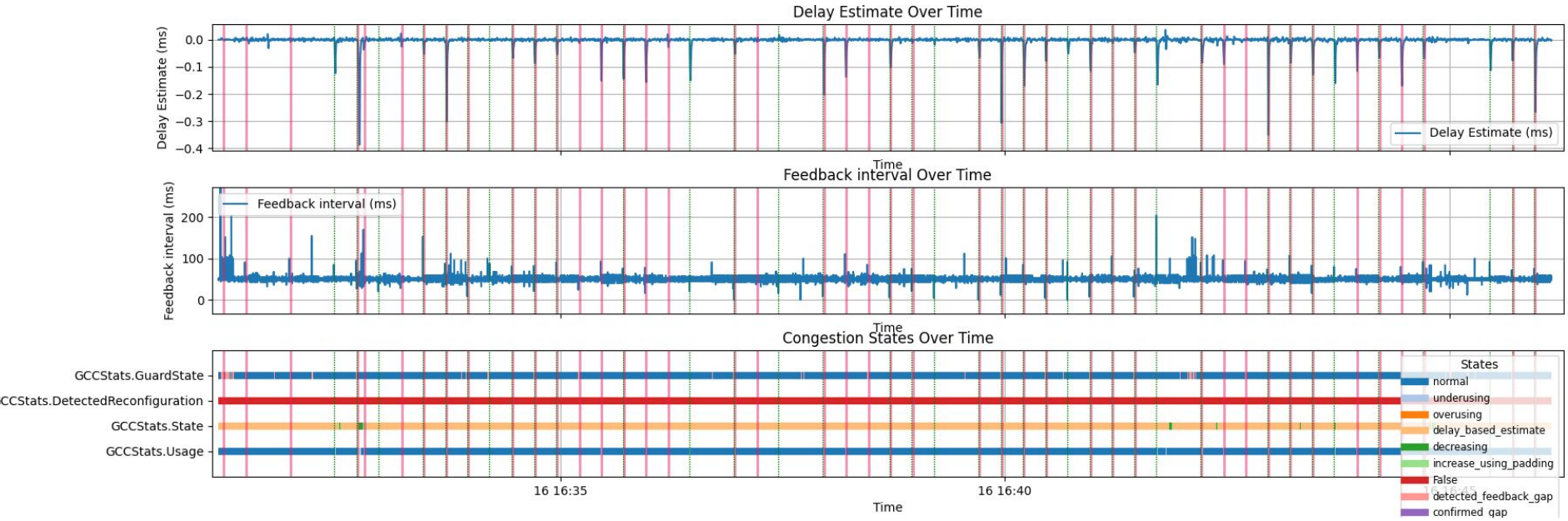
Visibility of known Reconfiguration Times

ICMP Pings and IRTT (10ms)



WebRTC RTT and Jitter

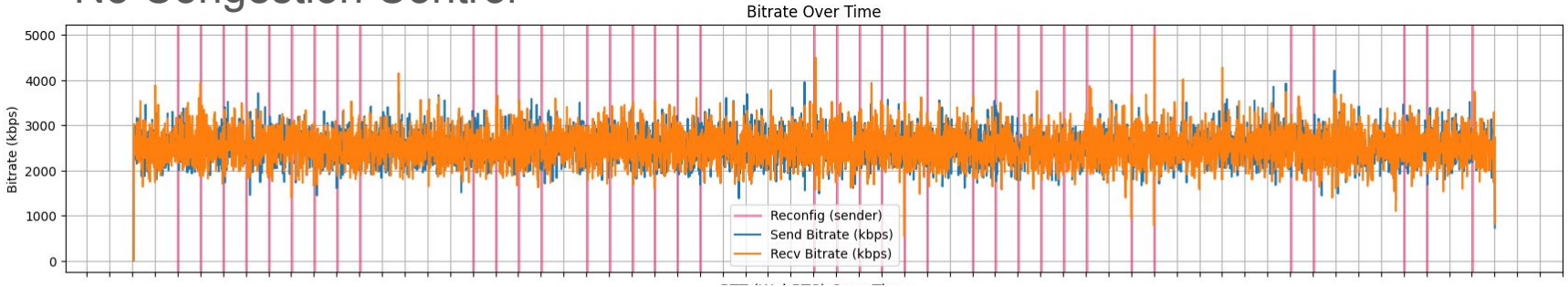
GCC Feedback Interval and Delay Estimate



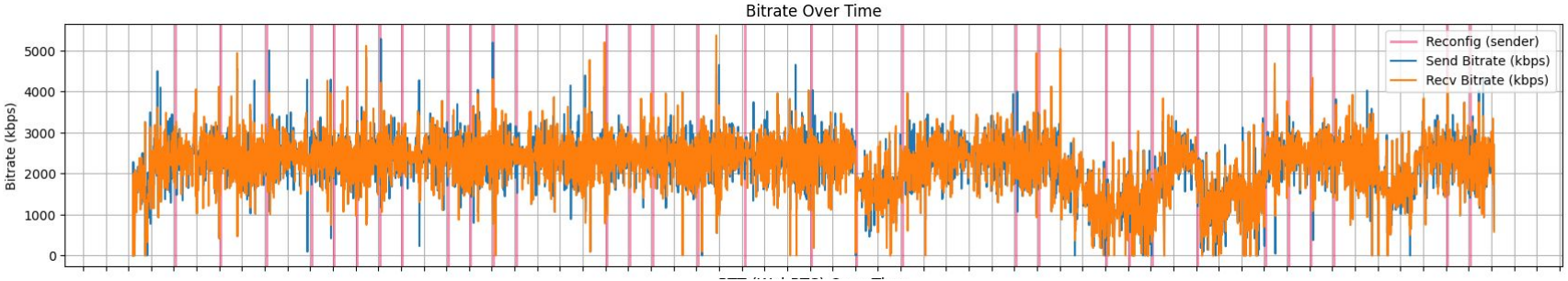
Impact of Reconfigurations

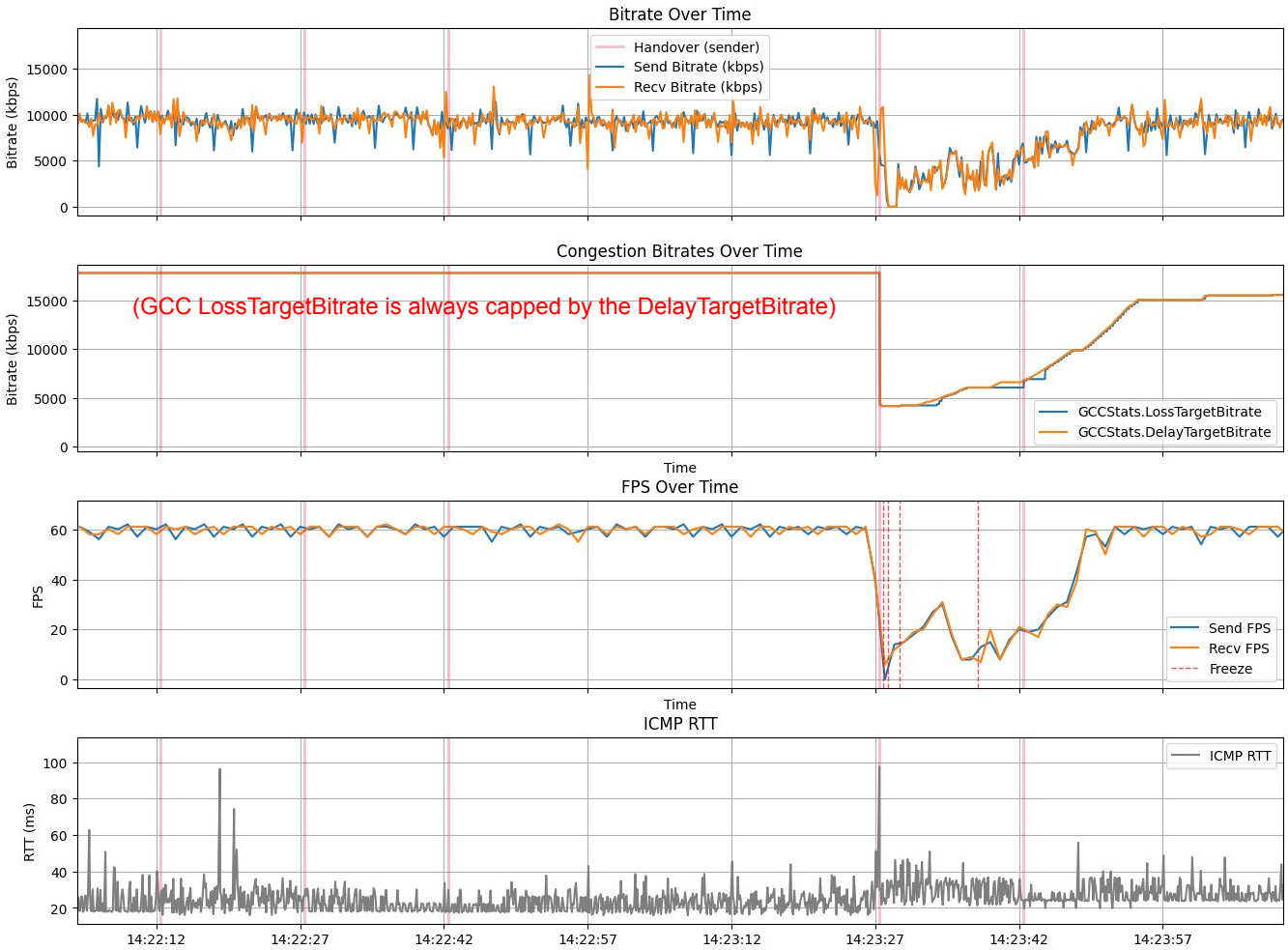
Collapse of Estimated Available Bitrate

No Congestion Control

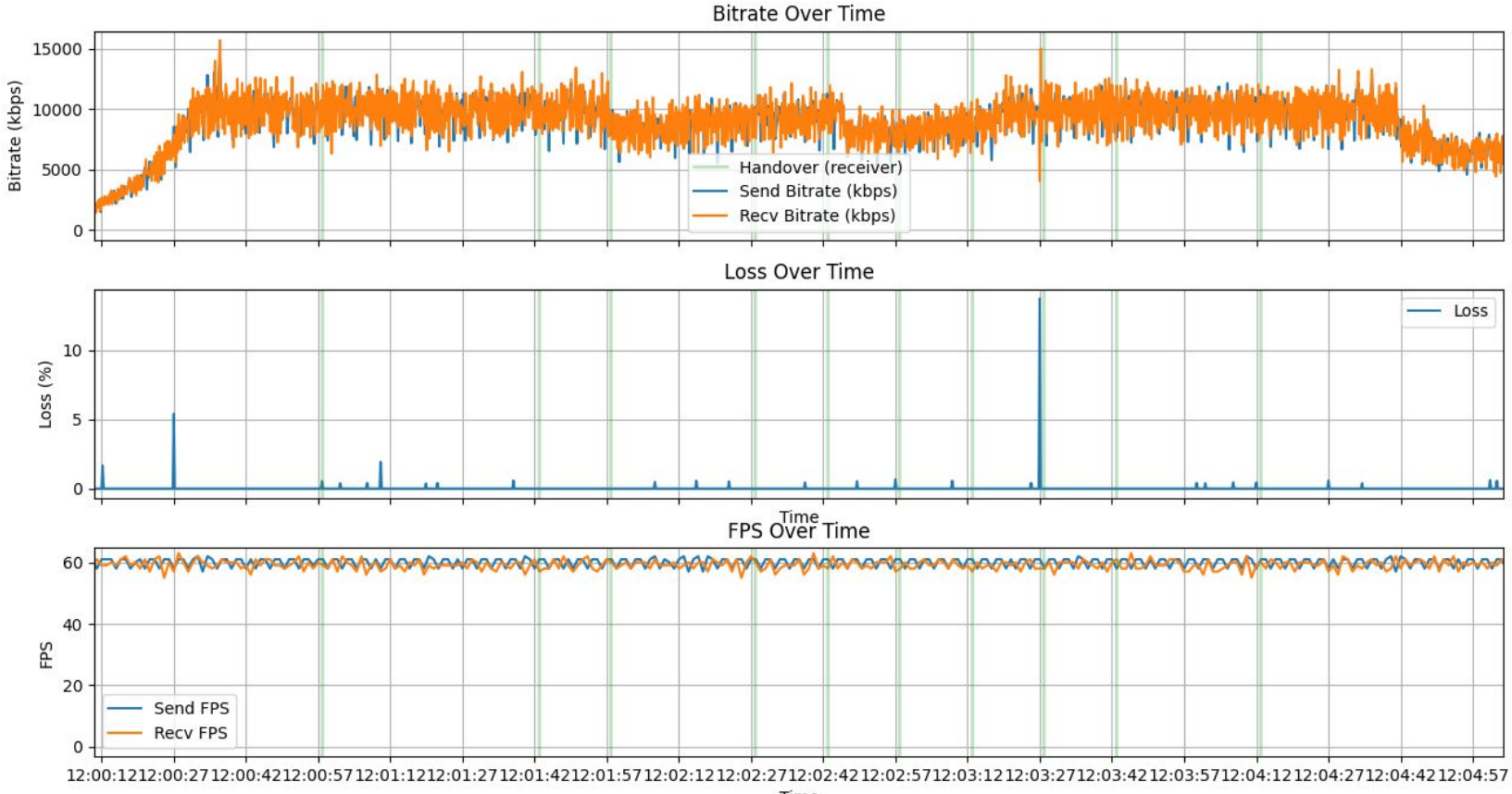


GCC




Bitrate Over Time
Handover (sender)
Send Bitrate (kbps)
Recv Bitrate (kbps)
Bitrate (kbps)
15000
10000
5000
0
Congestion Bitrates Over Time
(GCC LossTargetBitrate is always capped by the DelayTargetBitrate)
GCCStats.LossTargetBitrate
GCCStats.DelayTargetBitrate
Time
FPS Over Time
FPS
60
40
20
0
Send FPS
Recv FPS
Freeze
Time
ICMP RTT
ICMP RTT
RTT (ms)
100
80
60
40
20
14:22:12
14:22:27
14:22:42
14:22:57
14:23:12
14:23:27
14:23:42
14:23:57

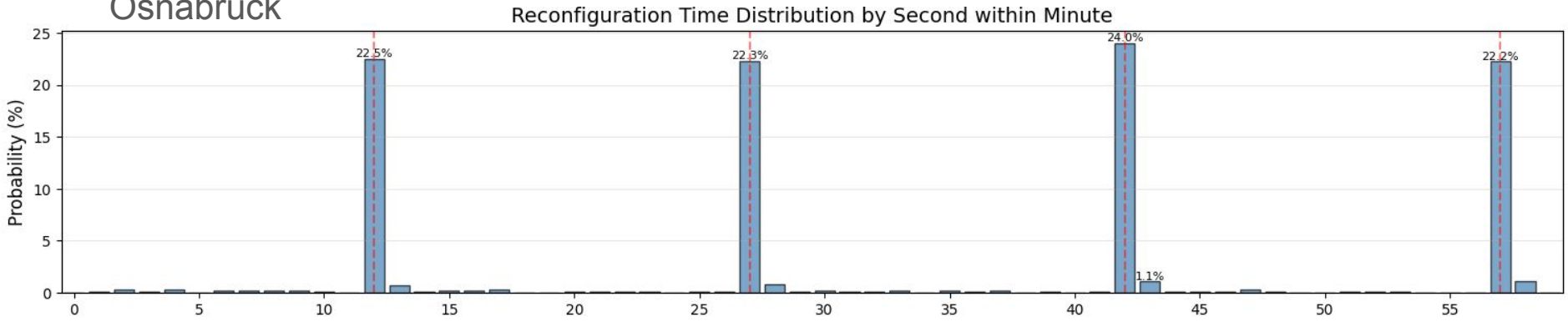
Negligible Impact on Downlink



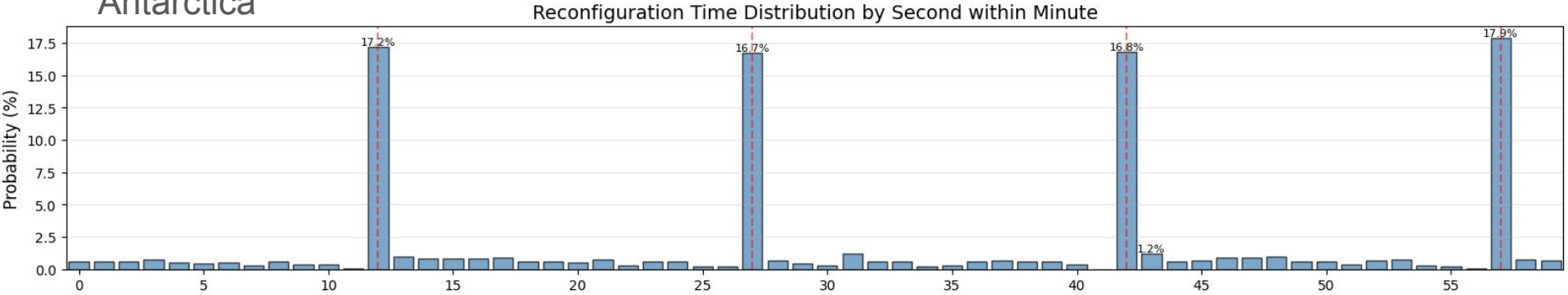
Irregular Handovers and Bad Paths

Percentage of Irregular Handovers

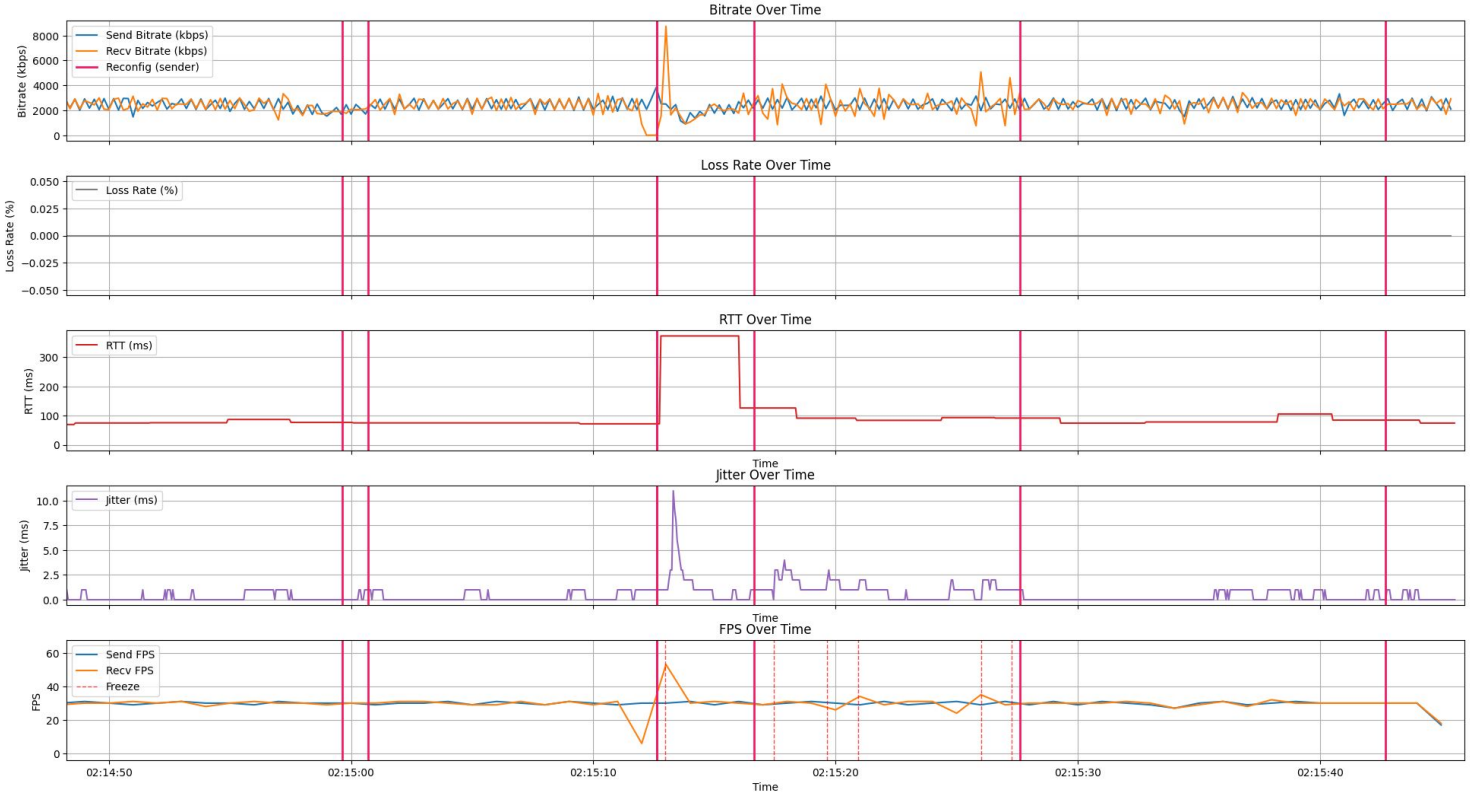
Osnabrück



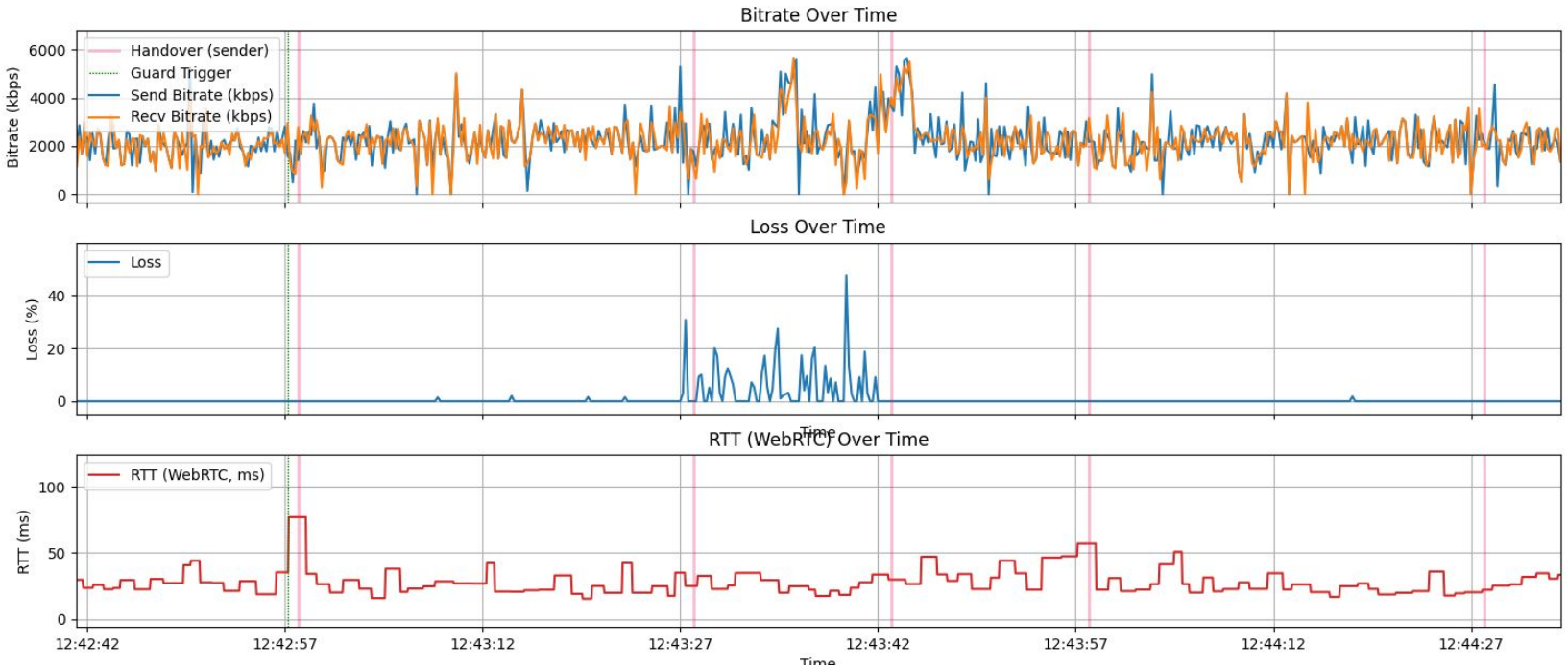
Antarctica



Irregular Handover Example



Bad Path Example

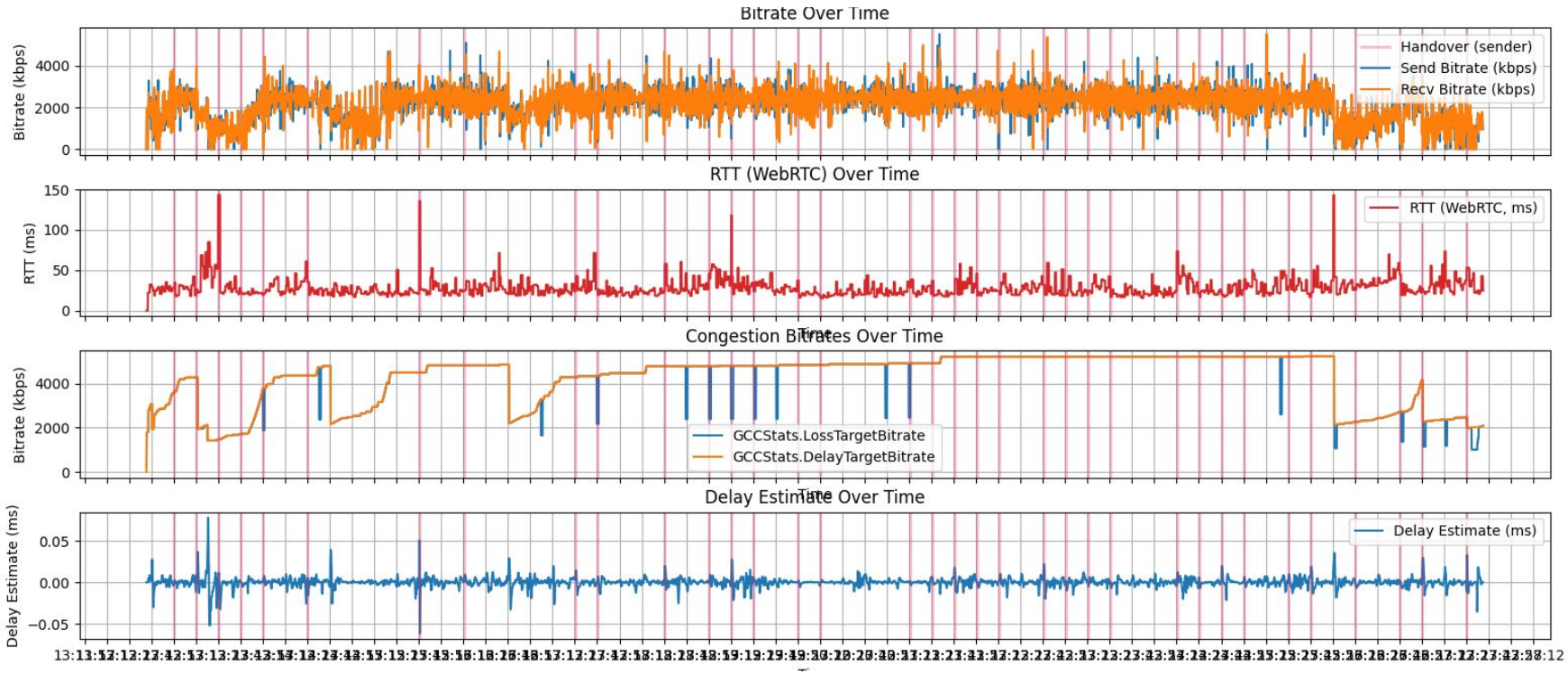


GCC Modifications

Predictive Reconfiguration Avoidance

- Reconfigurations happen at set 15-second interval
- Visible as spike in the delay estimate
- → Use autocorrelation on delay estimate history to predict next reconfiguration timestamp to apply predictive measures
  - Preemptively lower bitrate to reduce number of delayed or lost packets
  - Ignore some congestion signals during predicted period

Predictive Reconfiguration Avoidance

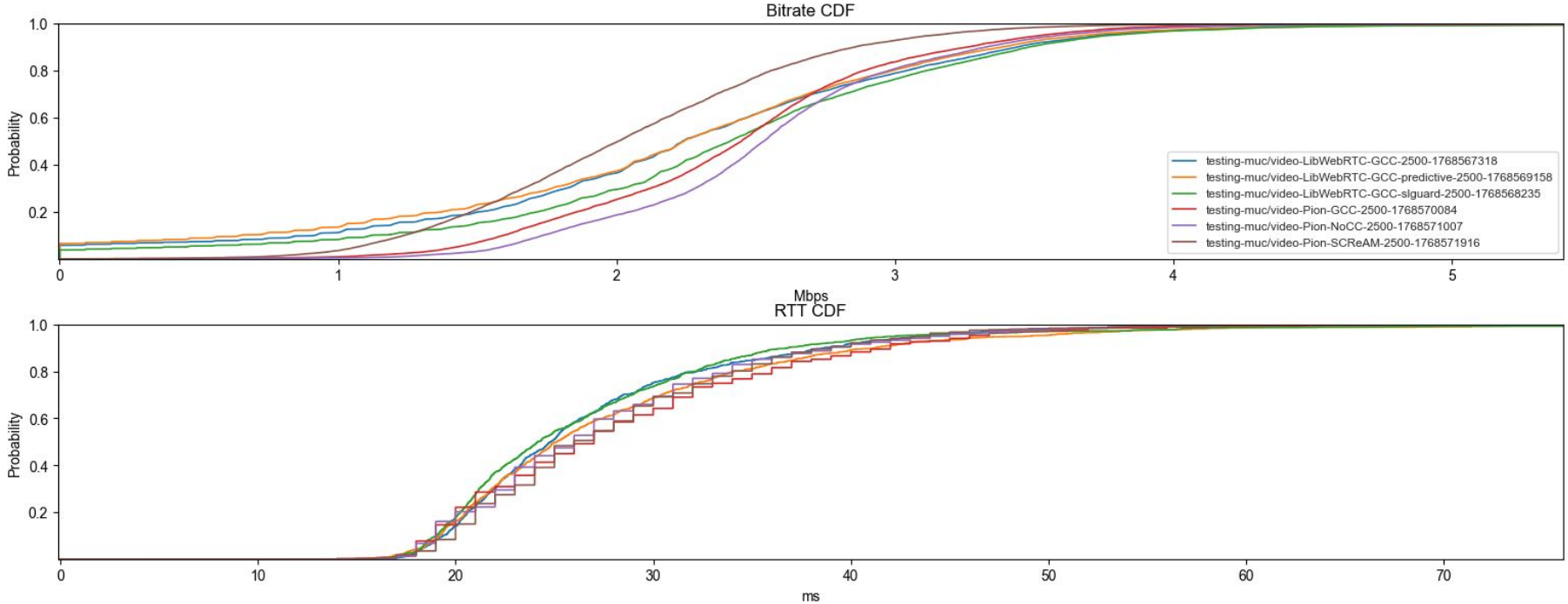


Reactive Reconfiguration Handling

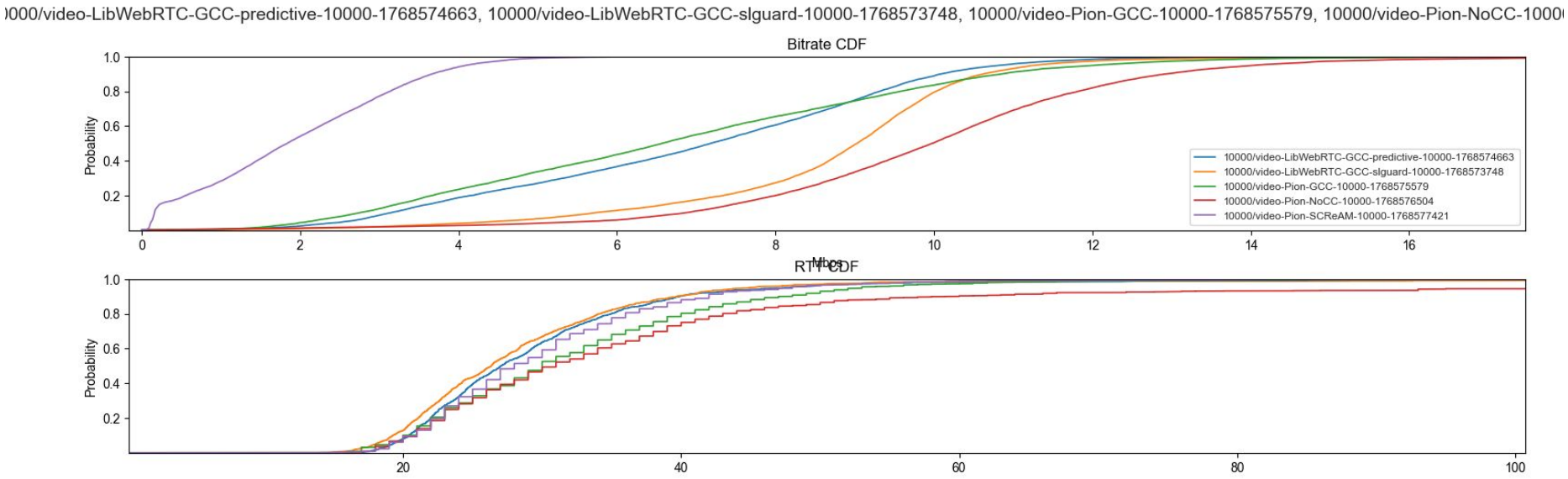
- Use Feedback Interval + Delay Estimate signal discussed earlier to catch reconfiguration after it happened
  - Separately they only show congestion or delay on a single path, Reconfigurations show minor delays in both directions
- Adjust GCC response to congestion signals to reduce impact
  - Ignore delay estimate updates until recovery
  - Potentially follow up with probes or application-level fast recovery

Comparison

768567318, testing-muc/video-LibWebRTC-GCC-predictive-2500-1768569158, testing-muc/video-LibWebRTC-GCC-slguard-2500-1768568235, testing-muc/video-Pion-GCC-2500-17685700

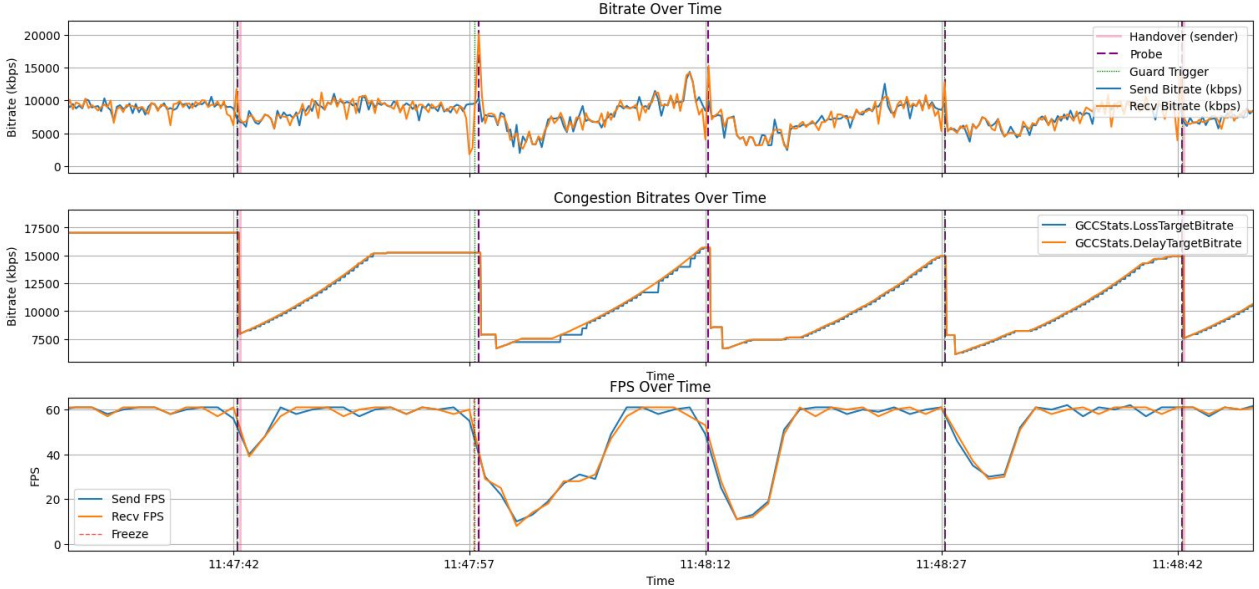


Comparison


)000/video-LibWebRTC-GCC-predictive-10000-1768574663, 10000/video-LibWebRTC-GCC-slguard-10000-1768573748, 10000/video-Pion-GCC-10000-1768575579, 10000/video-Pion-NoCC-1000
Bitrate CDF
Probability
Mbps
10000/video-LibWebRTC-GCC-predictive-10000-1768574663
10000/video-LibWebRTC-GCC-slguard-10000-1768573748
10000/video-Pion-GCC-10000-1768575579
10000/video-Pion-NoCC-10000-1768576504
10000/video-Pion-SCReAM-10000-1768577421
RTT CDF
Probability

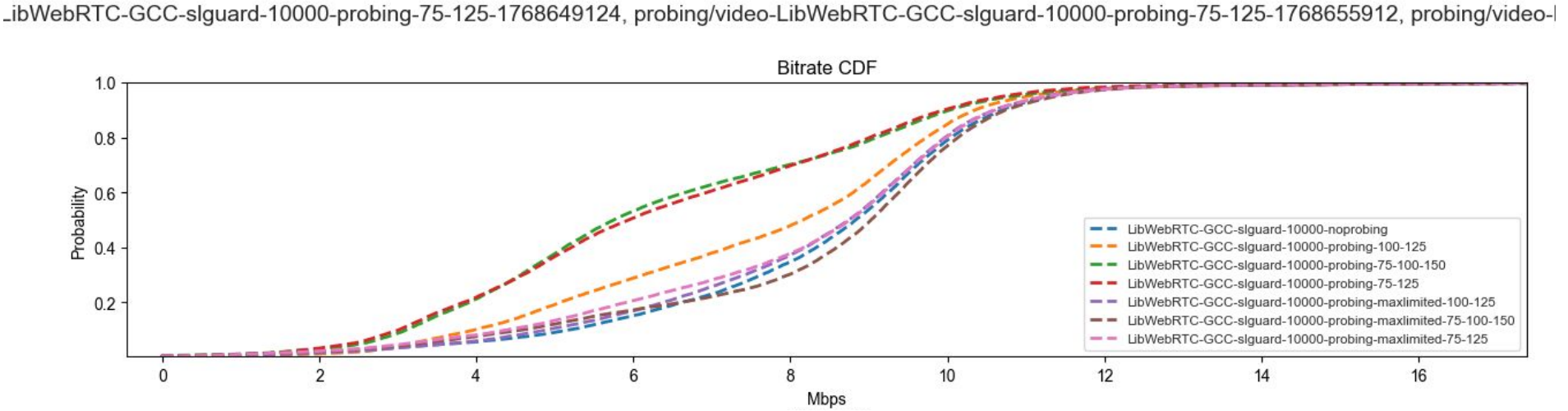
Reactive Reconfiguration Handling - Probing

- Probing traffic in addition to regular traffic can lead to congestion at high bitrates
- Example: 10k bitrate with probing set to 100% and 125% after reconfig

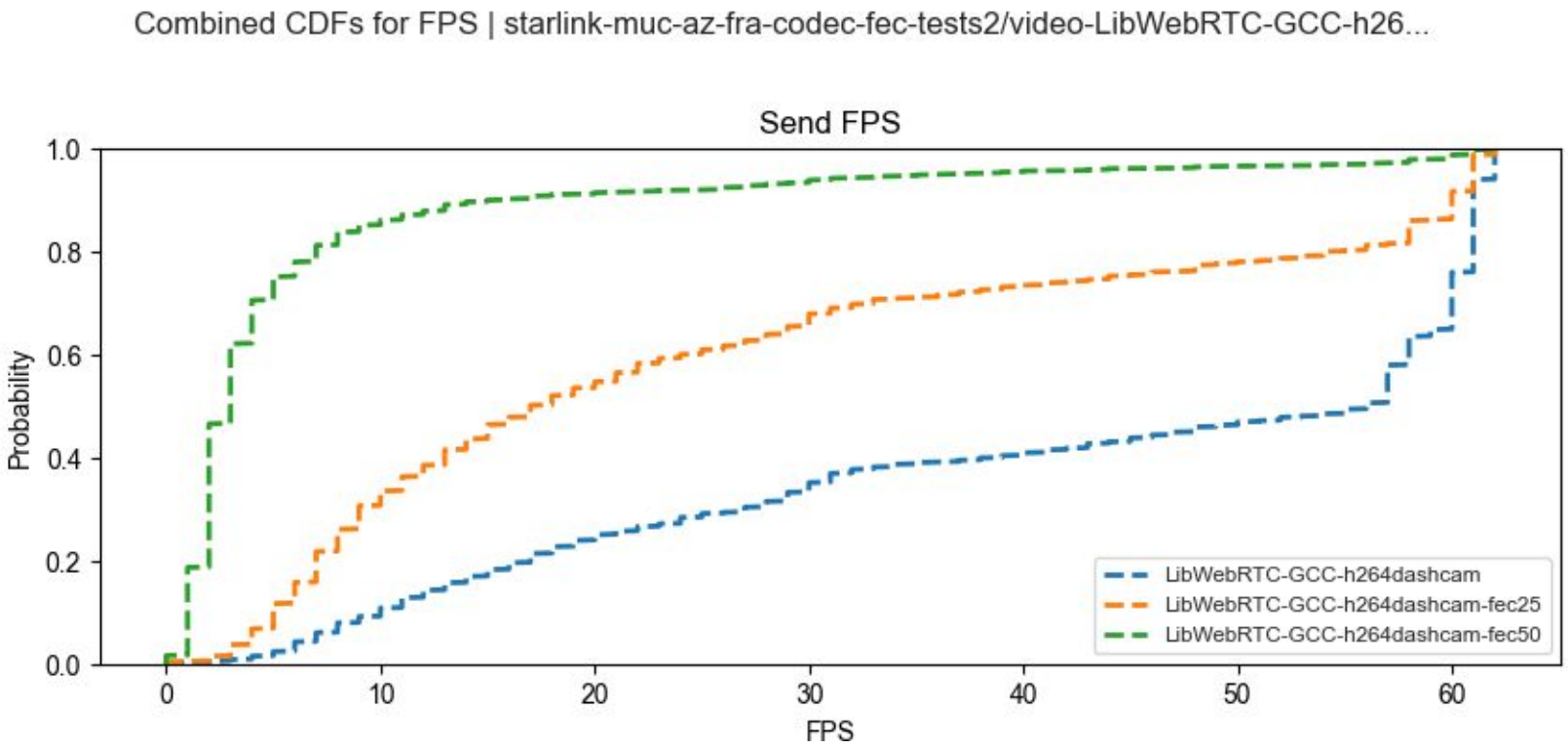


Reactive Reconfiguration Handling - Probing

- Probing only when falling below certain bitrate fixes this problem
- Impact of probing is negligible when this limit is enabled



Forced FEC


Combined CDFs for FPS | starlink-muc-az-fra-codec-fec-tests2/video-LibWebRTC-GCC-h26...
Send FPS
Probability
FPS
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
50
60
LibWebRTC-GCC-h264dashcam
LibWebRTC-GCC-h264dashcam-fec25
LibWebRTC-GCC-h264dashcam-fec50

Other Changes

- Increased jitter buffer size (200ms) has no effect on video quality (FPS / freezes)

- Automatically triggered intra-frame after detected reconfiguration has no effect on video quality (FPS / freezes)

Teams

Teams Findings

- Camera video feed sent with 2500kbps H.264
  - Low bitrate => fewer network issues/artifacts

- Meeting SFU based on location of first participant

- Traffic enters Azure network at closest Azure PoP
  - i.e. in a meeting hosted in Sydney with participant from Munich, the Munich participant will have traffic entering the Azure network in Frankfurt, after which it uses the Azure backbone
  - → SFU location has minor impact on QoE, just slightly higher latency